Lab 2: Data Carving

What You Need for this lab

- Install Virtualbox : https://www.virtualbox.org/wiki/Downloads
- Install Kali 2021.4. : https://old.kali.org/kali-images/kali-2021.4/
  - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- Run a tool installation script instructions, or you can simply follow the commands below : (the script ONLY is tested on Kali 2021.4)
  - wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Help/tool-install-zsh.sh
  - chmod +x tool-install-zsh.sh
  - ./tool-install-zsh.sh

Example scenarios

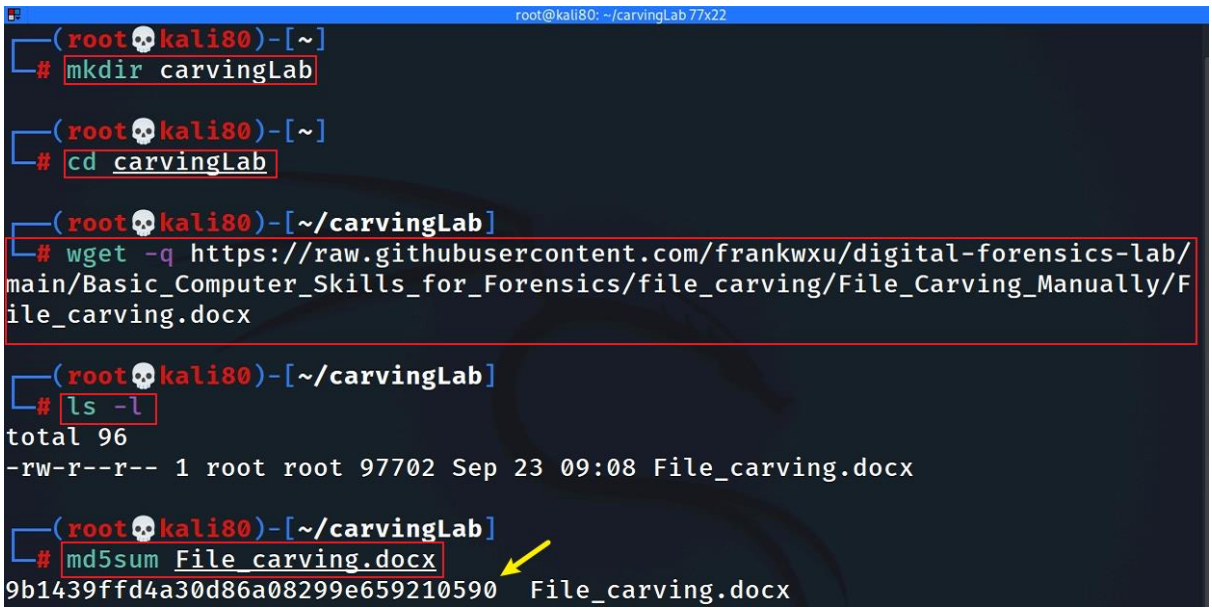
- Scenario 1: A file (A) is hidden inside of another file (B). You can't open the file B because the B's header is corrupted.
- Scenario 2: A suspect deleted files. The files contains an important information. A file occupies a few clusters. Unfortunately, some clusters are reused (overwritten) by new files.

A forensic expert really wants to recover files, even a partial files.

1. **Extracting images from a corrupted Word document**

Step 1.

- Prepare required files



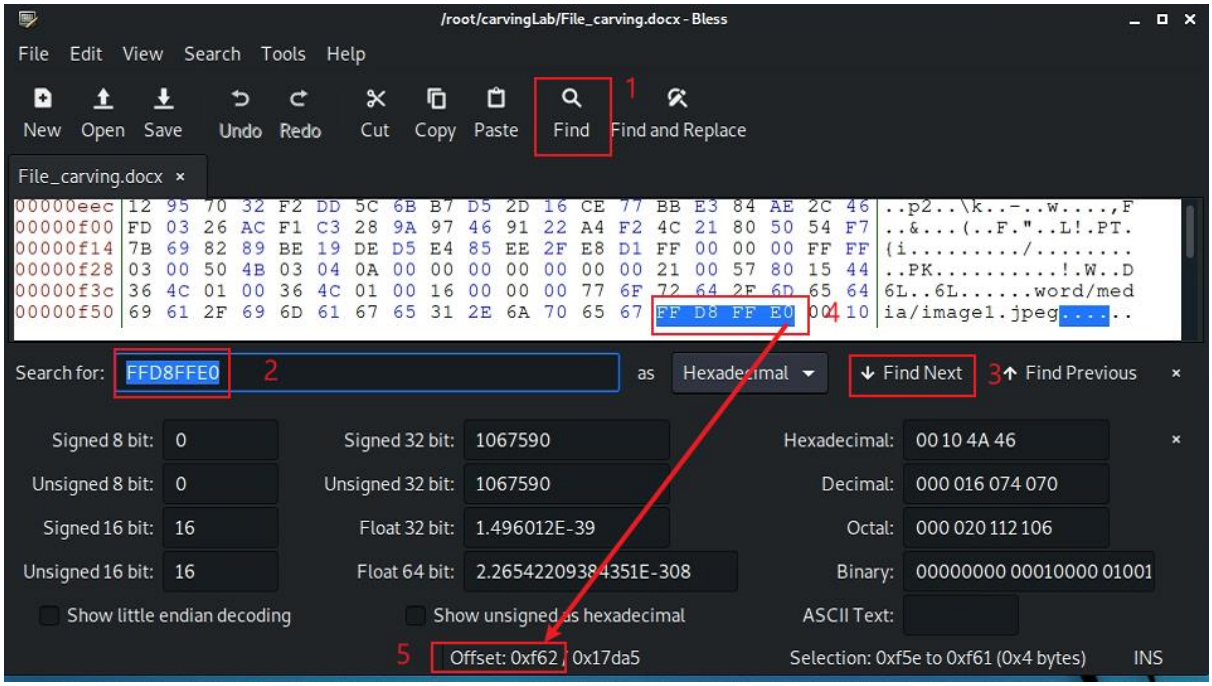
Step 2.

– View the file in a hex editor

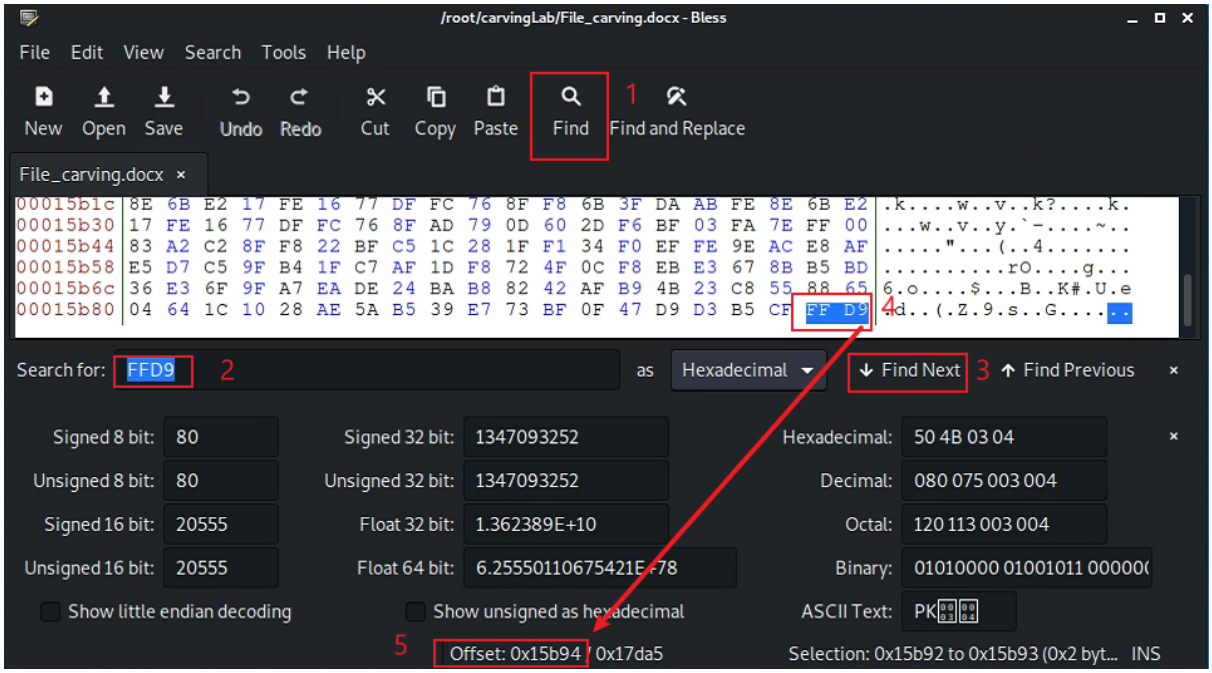
```
┌──(root💀kali80)-[~/carvingLab]
└─# apt-get install bless
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
bless is already the newest version (0.6.0-7).
0 upgraded, 0 newly installed, 0 to remove and 522 not upgraded.

┌──(root💀kali80)-[~/carvingLab]
└─# bless File_carving.docx
```

Step 3.

– Search file header start offset – 0F5E

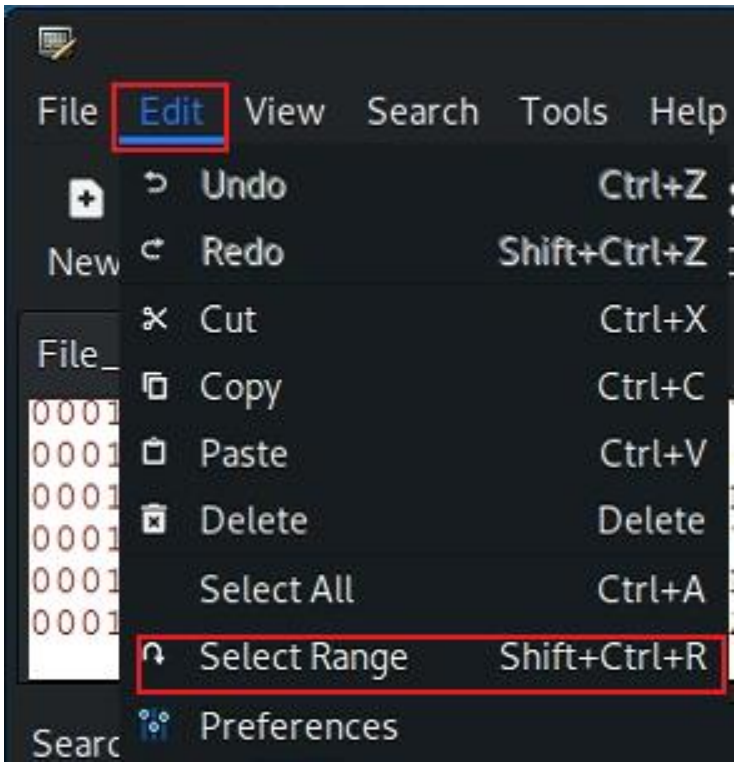


– Search file trailer ends offset – 15B93

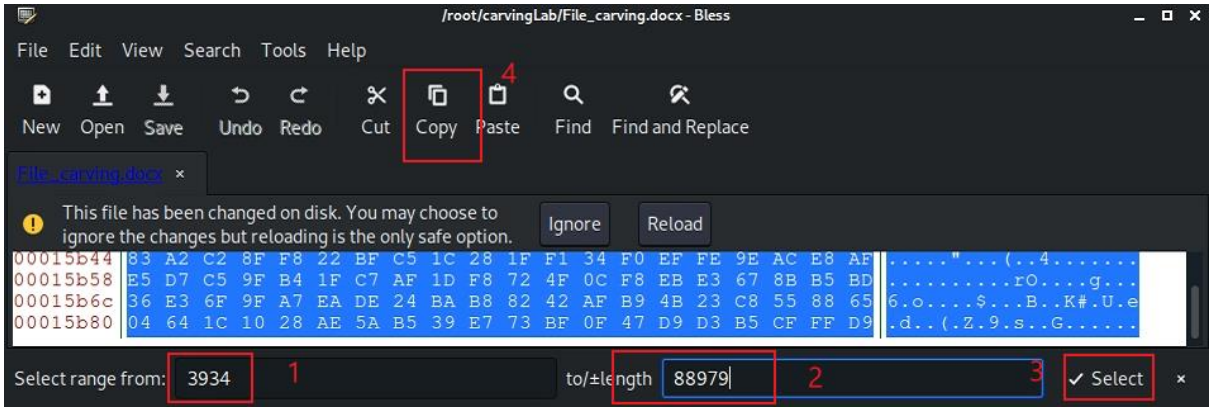


– Select hex from header to tail

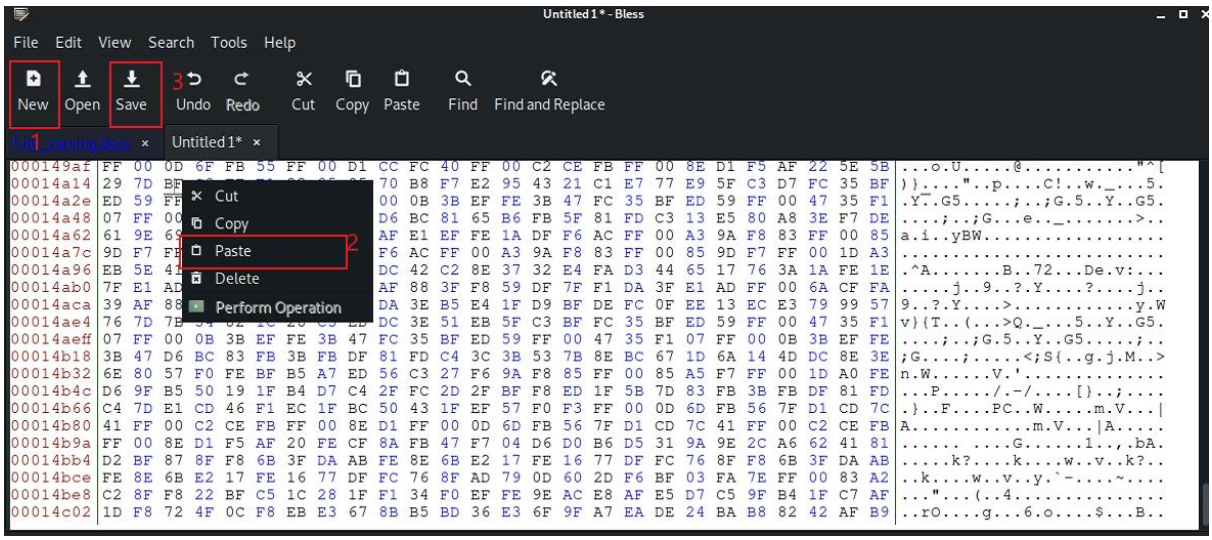
  - (0F5E)16=(3934)10
  - (15B93)16=(88979)10



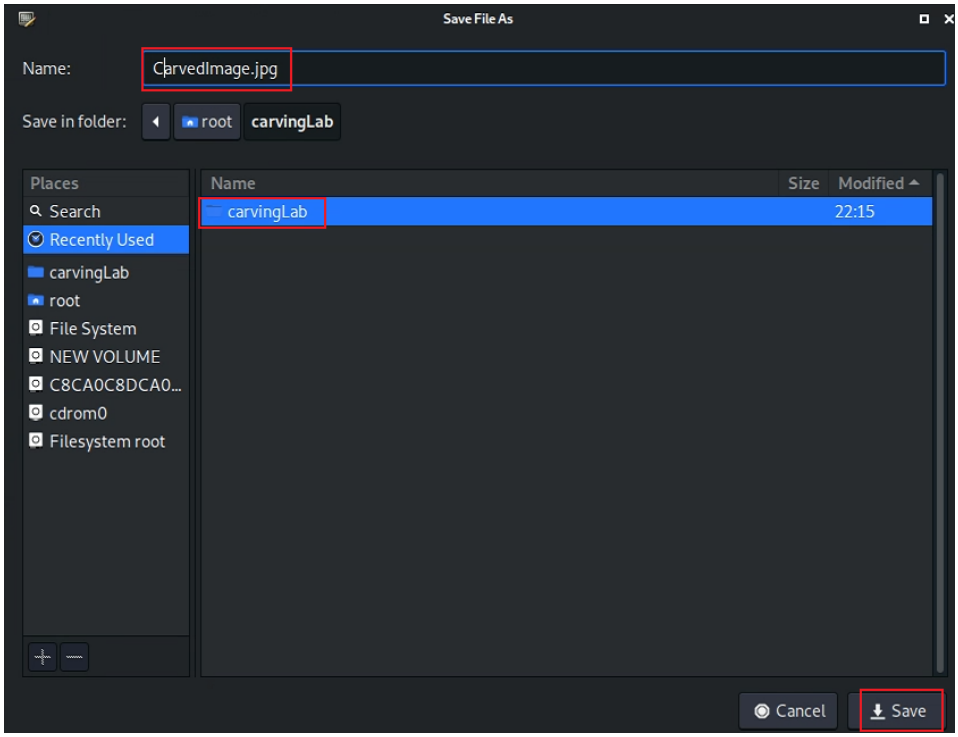
– Copy the selection



– Paste the selection

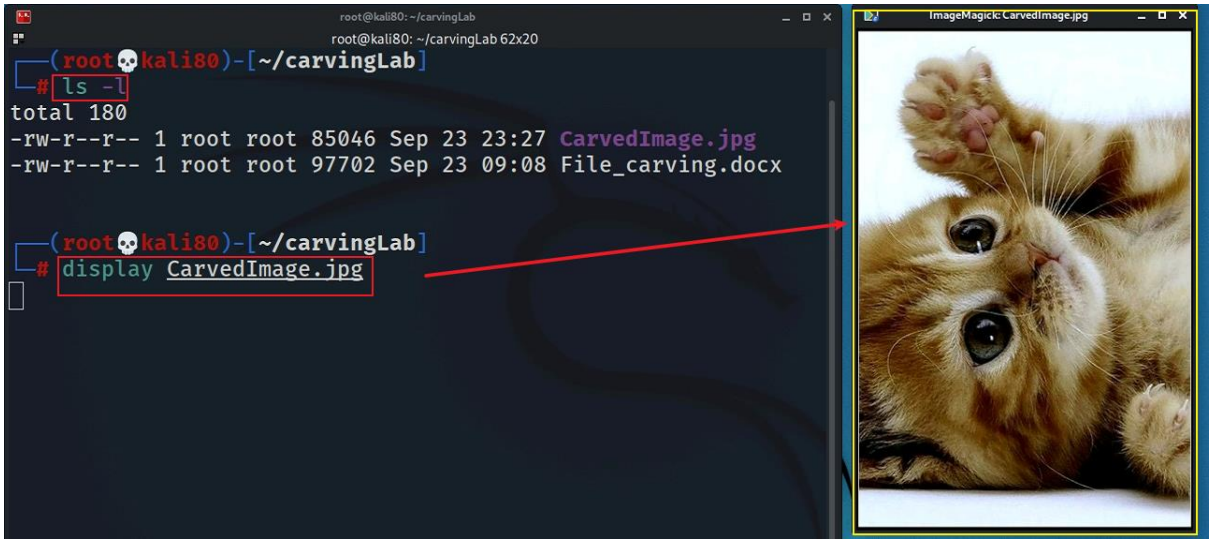


– Save the image



Step 4.

– Show the carved image



2. Carving/Recovering a USB image

– Prepare a USB image for file carving

Step 1.

– Download the zipped USB image

```
┌──(root💀kali80)-[~/carvingLab]
└─# wget  -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic
_Computer_Skills_for_Forensics/file_carving/usb_image/120M.7z

┌──(root💀kali80)-[~/carvingLab]
└─# ls -l 120M.7z
-rw-r--r-- 1 root root 36720470 Sep 24 09:03 120M.7z
```

– Compute hashes

```
┌──(root💀kali80)-[~/carvingLab]
└─# hashdeep -c md5,sha1 120M.7z
%%%% HASHDEEP-1.0
%%%% size,md5,sha1,filename
## Invoked from: /root/carvingLab
## # hashdeep -c md5,sha1 120M.7z
##
36720470,dfe7b5424e54cd1bf50d5df47aceeb3c,2810745018afaa2da31dc17a8eb590fca66eeef7,/root
/carvingLab/120M.7z
```

(size, md5, sha1)

– List the content of the zipped file

```
┌──(root💀kali80)-[~/carvingLab]
└─# 7z l 120M.7z

7-Zip [64] 16.02 : Copyright (c) 1999-2016 Igor Pavlov : 2016-05-21
p7zip Version 16.02 (locale=en_US.UTF-8,Utf16=on,HugeFiles=on,64 bits,2 CPUs Intel(R) Xe
on(R) Gold 6226R CPU @ 2.90GHz (50650),ASM,AES-NI)

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Listing archive: 120M.7z

--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

   Date      Time    Attr         Size   Compressed  Name
------------------- ----- ------------ ------------  ------------------------
2021-09-22 22:59:31 D...A            0            0  120M
2021-09-22 22:59:31 ....A    124780544     36720256  120M/usb_fat_carving.001
2021-09-22 22:59:32 ....A         1685               120M/usb_fat_carving.001.txt
------------------- ----- ------------ ------------  ------------------------
2021-09-22 22:59:32          124782229     36720256  2 files, 1 folders
```

content of the zip

– List the content of the zipped file

```
┌──(root💀kali80)-[~/carvingLab]
└─# 7z e 120M.7z

7-Zip [64] 16.02 : Copyright (c) 1999-2016 Igor Pavlov : 2016-05-21
p7zip Version 16.02 (locale=en_US.UTF-8,Utf16=on,HugeFiles=on,64 bits,2 CPUs Intel(R) Xe
on(R) Gold 6226R CPU @ 2.90GHz (50650),ASM,AES-NI)

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size:       124782229
Compressed: 36720470
```

– Verify the hashes

```
┌──(root💀kali80)-[~/carvingLab]
└─# hashdeep -c md5,sha1 usb_fat_carving.001
%%%% HASHDEEP-1.0
%%%% size,md5,sha1,filename
## Invoked from: /root/carvingLab
## # hashdeep -c md5,sha1 usb_fat_carving.001
##
124780544,ba4a1d0ba49f4a6667b00a3b3e85e604,bcc2d49fd49c9521ecb1739f6542c6bf327375ef,/roo
t/carvingLab/usb_fat_carving.001

┌──(root💀kali80)-[~/carvingLab]
└─# cat usb_fat_carving.001.txt | grep "checksum"
MD5 checksum:    ba4a1d0ba49f4a6667b00a3b3e85e604
SHA1 checksum:   bcc2d49fd49c9521ecb1739f6542c6bf327375ef
MD5 checksum:    ba4a1d0ba49f4a6667b00a3b3e85e604 : verified
SHA1 checksum:   bcc2d49fd49c9521ecb1739f6542c6bf327375ef : verified
```

Step 2.

- Exam the content of the USB
- Display partitions

```
┌──(root💀kali80)-[~/carvingLab]
└─# fdisk -l usb_fat_carving.001
Disk usb_fat_carving.001: 119 MiB, 124780544 bytes, 243712 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xa1159a00

Device                 Boot Start    End Sectors   Size Id Type
usb_fat_carving.001p1 *      128 243711  243584 118.9M  e W95 FAT16 (LBA)
```

- Find deleted files

```
┌──(root💀kali80)-[~/carvingLab]
└─# fls -o 128 usb_fat_carving.001
r/r 3:  USB         (Volume Label Entry)
d/d 6:  System Volume Information
r/r * 7:        _est
r/r 10: .dropbox.device
d/d * 13:       old_File_Carving_files
r/r * 15:       B_ub_poe4.bmp
r/r * 18:       B_zoom-eubie-mono.bmp
r/r * 21:       Ballardlab8.java
r/r * 24:       brittLab10.java
r/r * 27:       DO_example.doc
r/r * 30:       DO_example2.doc
r/r * 33:       G_BuiltForThis.gif
r/r * 35:       G_zoom-sc.gif
r/r * 37:       H_Form.html
r/r * 39:       H_hello.html
r/r * 41:       J_ub_law.jpg
r/r * 44:       J_ub_night.jpg
r/r * 47:       nps-2008-jean_outlook.pst
r/r * 50:       P_CAS-zoom-6.png
r/r * 53:       P_MSB_1_zoom.png
r/r * 57:       pd_Evidence_search_techniques.pdf
r/r * 61:       pd_Forensic_Report_Template.pdf
r/r * 64:       pp_Number_Systems.pptx
r/r * 67:       pp_one_page.pptx
r/r 69: readme.docx
r/r 70: readme.txt
r/r * 71:       _tf_1.rtf
r/r * 74:       T_Eubie-iphone-5-8.tiff
r/r * 78:       T_youknowus-iphone-5-8.tiff
r/r * 81:       W_axloadcomplete.wav
r/r * 84:       W_speaker_test_sound.wav
r/r * 86:       Z_file.zip
r/r * 88:       江莎莎行踪轨迹.doc
r/r * 91:       nps-2008-jean_outlook.pst
v/v 3889667:    $MBR
v/v 3889668:    $FAT1
v/v 3889669:    $FAT2
V/V 3889670:    $OrphanFiles
```

deleted

deleted

- Decide which file types need to carve

```
┌──(root💀kali80)-[~/carvingLab]
└─# leafpad /etc/scalpel/scalpel.conf
/usr/share/themes/Kali-Dark/gtk-2.0/gtkrc:39: Unable to find include file: "apps.rc"
/usr/share/themes/Kali-Dark/gtk-2.0/gtkrc:40: Unable to find include file: "hacks.rc"
/usr/share/themes/Kali-Dark/gtk-2.0/gtkrc:41: Unable to find include file: "hacks-dark.r
c"
```

*scalpel.conf

File Edit Search Options Help

```
#----------------------------------------------------------------------
#
# Here is an example of how to use the no extension option. Any files
# beginning with the string "FOREMOST" are carved and no file extensions
# are used. No footer is defined and the max carve size is 1000 bytes.
#
#      NONE     y      1000     FOREMOST
#
#----------------------------------------------------------------------
# GRAPHICS FILES
#----------------------------------------------------------------------
#
#
# AOL ART files
#       art     y       150000  \x4a\x47\x04\x0e        \xcf\xc7\xcb
#       art     y       150000  \x4a\x47\x03\x0e        \xd0\xcb\x00\x00
#
# GIF and JPG files (very common)
#       gif     y       5000000         \x47\x49\x46\x38\x37\x61        \x00\x3b
#       gif     y       5000000         \x47\x49\x46\x38\x39\x61        \x00\x3b
        jpg     y       5242880         \xff\xd8\xff???Exif             \xff\xd9        REVERSE
        jpg     y       5242880         \xff\xd8\xff???JFIF             \xff\xd9        REVERSE
#
#
```

uncomment #

- Save it and quit!
- Show help

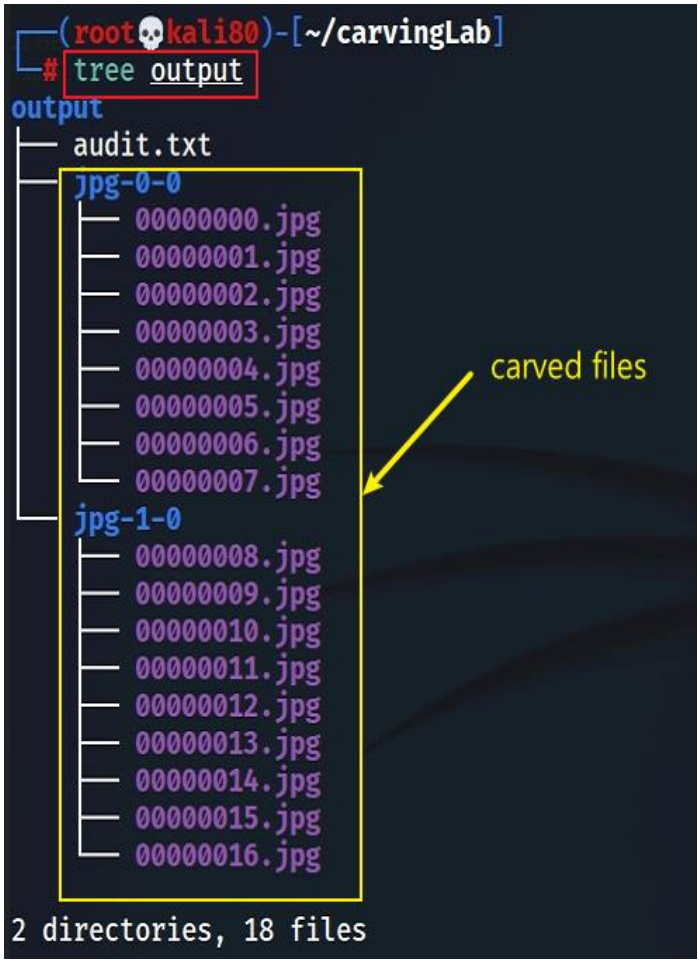
```
┌──(root💀kali80)-[~/carvingLab]
└─# scalpel -h
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.
Carves files from a disk image based on file headers and footers.

Usage: scalpel [-b] [-c <config file>] [-d] [-h|V] [-i <file>]
               [-m blocksize] [-n] [-o <outputdir>] [-O num] [-q clustersize]
               [-r] [-s num] [-t <blockmap file>] [-u] [-v]
               <imgfile> [<imgfile>] ...

-b  Carve files even if defined footers aren't discovered within
    maximum carve size for file type [foremost 0.69 compat mode].
-c  Choose configuration file.
-d  Generate header/footer database; will bypass certain optimizations
    and discover all footers, so performance suffers.  Doesn't affect
    the set of files carved.  **EXPERIMENTAL**
-h  Print this help message and exit.
-i  Read names of disk images from specified file.
-m  Generate/update carve coverage blockmap file.  The first 32bit
    unsigned int in the file identifies the block size. Thereafter
    each 32bit unsigned int entry in the blockmap file corresponds
    to one block in the image file.  Each entry counts how many
    carved files contain this block. Requires more memory and
    disk.  **EXPERIMENTAL**
-n  Don't add extensions to extracted files.
-o  Set output directory for carved files.
-O  Don't organize carved files by type. Default is to organize carved files
    into subdirectories.
-p  Perform image file preview; audit log indicates which files
    would have been carved, but no files are actually carved.
-q  Carve only when header is cluster-aligned.
-r  Find only first of overlapping headers/footers [foremost 0.69 compat mode].
-s  Skip n bytes in each disk image before carving.
-t  Set directory for coverage blockmap.  **EXPERIMENTAL**
-u  Use carve coverage blockmap when carving.  Carve only sections
    of the image whose entries in the blockmap are 0.  These areas
    are treated as contiguous regions.  **EXPERIMENTAL**
```

Step 3.

– Carving the USB image

```
┌──(root💀kali80)-[~/carvingLab]
└─# scalpel usb_fat_carving.001 -o output                                    1 ×
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/root/carvingLab/usb_fat_carving.001"

Image file pass 1/2.
usb_fat_carving.001: 100.0% |**********************************|  119.0 MB    00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built.  Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" --> 8 f
iles
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" --> 9 f
iles
Carving files from image.
Image file pass 2/2.
usb_fat_carving.001: 100.0% |**********************************|  119.0 MB    00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 0 seconds.
```

– Show carved files

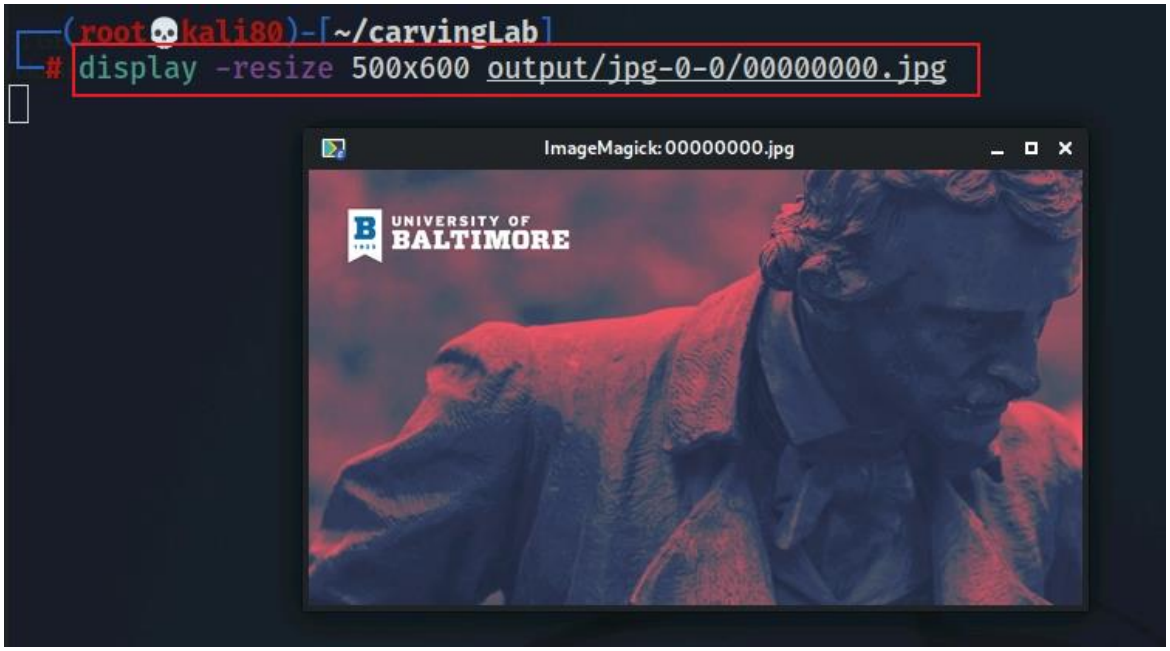


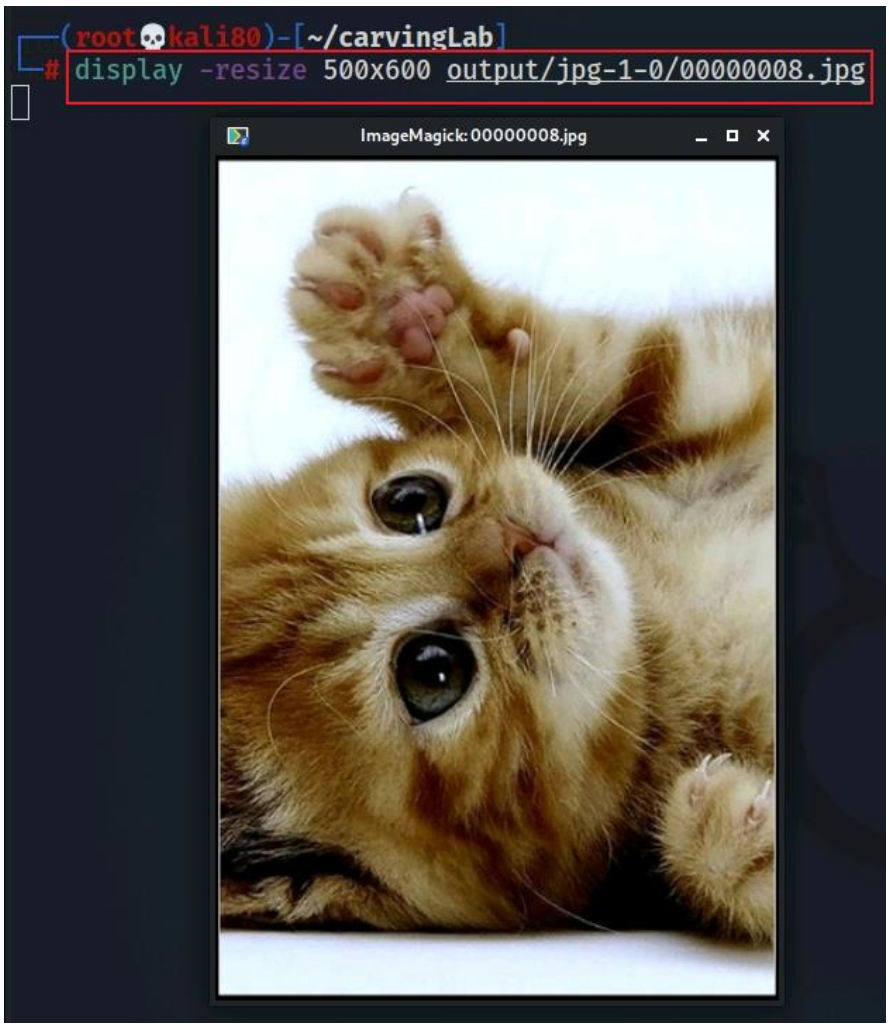
– Show audit log

```
(root kali80)-[~/carvingLab]
# cat output/audit.txt

Scalpel version 1.60 audit file
Started at Fri Sep 24 11:20:43 2021
Command line:
scalpel usb_fat_carving.001 -o output

Output directory: /root/carvingLab/output
Configuration file: /etc/scalpel/scalpel.conf

Opening target "H="

The following files were carved:
File              Start              Chop          Length          Extracted From
00000010.jpg      3796992            NO           3148500          usb_fat_carving.001
00000009.jpg       425822            NO           4875802          usb_fat_carving.001
00000008.jpg       335872            NO           4965752          usb_fat_carving.001
00000002.jpg      6938624            NO              6868          usb_fat_carving.001
00000001.jpg      5285888            NO           1659604          usb_fat_carving.001
00000000.jpg      3897344            NO           3048148          usb_fat_carving.001
00000004.jpg     15427584            NO           4223822          usb_fat_carving.001
00000003.jpg     12881920            NO           4256310          usb_fat_carving.001
00000006.jpg     19638272            NO           5229050          usb_fat_carving.001
00000005.jpg     17121280            NO           4248530          usb_fat_carving.001
00000007.jpg     21358592            NO           5183267          usb_fat_carving.001
00000016.jpg     36671488            NO           2889262          usb_fat_carving.001
00000015.jpg     36571136            NO           2989614          usb_fat_carving.001
00000014.jpg     36393610            NO           3167140          usb_fat_carving.001
00000013.jpg     36352448            NO           3208302          usb_fat_carving.001
00000012.jpg     35590996            NO           3969754          usb_fat_carving.001
00000011.jpg     35350242            NO           4210508          usb_fat_carving.001


Completed at Fri Sep 24 11:20:43 2021
```

Step 4.

– Display two carved jpg image





YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 2.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 2 From YOUR NAME**", replacing "YOUR NAME" with your real name.